

Master Maths Worksheet 51

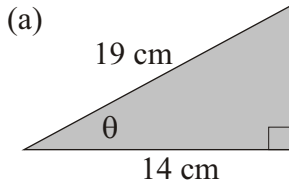
Trigonometry - Calculating Angles

51

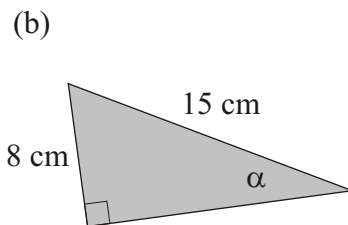
Name: _____

Give all answers correct to one decimal place.

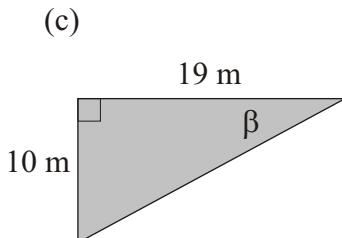
1. Find the unknown angles in the following triangles.



$\theta =$

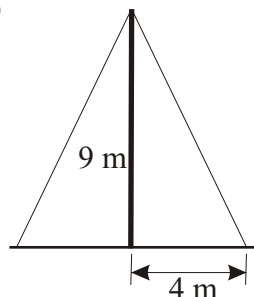


$\alpha =$



$\beta =$

2. Cables fixed to the top of a 9 m pole are fixed to the ground 4 m from the pole. What angle do these cables make with the ground?



$\theta =$

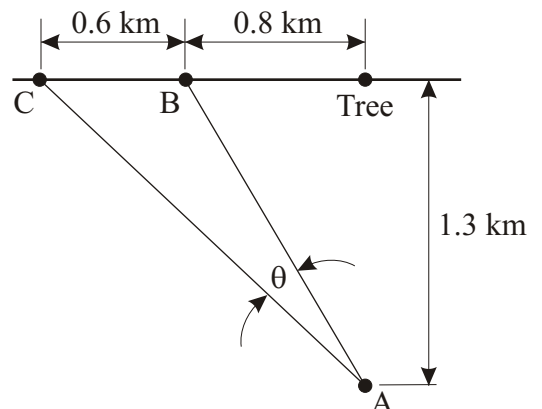
3. Two hikers walk 5 km south from their camp then walk 3 km east.

(a) Draw a diagram showing the path walked by the hikers.

(b) Calculate the direction the hikers need to take to walk in a straight line back to their camp.

4. A cameraman is positioned at point A which is 1.3 km south of a tree as shown on the diagram below. He spots an elephant at point B which is 0.8 km west of the tree. He films it as it moves to point C, 0.6 km from B.

Through what angle, θ , does he need to move his camera?



$\theta =$